Deck A · Energy and ATP

Unit 3 · Energy and Matter
12 cards + this cover
Label code **Bio · U3 · L1 · A**
Envelope: **U3-A** · Set ____ of ____
Print single-sided on yellow cardstock. Count against this card before the set goes back in the bin.
Key: OneDrive › Unit 03 › Answer Keys

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Immediate energy

Bio · U3 · L1 · A · 04/12

Short-term storage

Bio · U3 · L1 · A · 09/12

Long-term storage

Bio · U3 · L1 · A · 06/12

ATP

adenine, a sugar and **three** phosphate groups: the charged form

Bio · U3 · L1 · A · 10/12

ATP

spent within seconds, wherever work is done

Bio · U3 · L1 · A · 01/12

ATP again: rebuilt, not used up. Back to the start.

Bio · U3 · L1 · A · 05/12

Glucose is broken down in the mitochondrion (cellular respiration)

Bio · U3 · L1 · A · 02/12

Energy is released for cell work: a pump, a muscle, building a protein

Bio · U3 · L1 · A · 08/12

ADP + phosphate

two phosphate groups left: the spent form

Bio · U3 · L1 · A · 07/12

That energy reattaches the third phosphate

Bio · U3 · L1 · A · 12/12

Fat (a lipid)

packed away for weeks or months

Bio · U3 · L1 · A · 03/12

Glucose

a sugar carried in the blood or made in a leaf

Bio · U3 · L1 · A · 11/12

Deck B · Photosynthesis or Respiration?

Unit 3 · Energy and Matter
24 cards + this cover
Label code **Bio · U3 · L1 · B**
Envelope: **U3-B** · Set ____ of ____
Print single-sided on yellow cardstock. Count against this card before the set goes back in the bin.
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Photosynthesis (chloroplast)

Bio · U3 · L1 · B · 01/24

Cellular respiration (mitochondrion)

Bio · U3 · L1 · B · 20/24

Both

Bio · U3 · L1 · B · 06/24

Oxygen goes in

Bio · U3 · L1 · B · 18/24

Plants

Bio · U3 · L1 · B · 03/24

Water comes out

Bio · U3 · L1 · B · 19/24

Light energy goes in

Bio · U3 · L1 · B · 17/24

Algae

Bio · U3 · L1 · B · 05/24

Carbon dioxide goes in

Bio · U3 · L1 · B · 02/24

Carbon dioxide comes out

Bio · U3 · L1 · B · 21/24

Cyanobacteria
no chloroplasts and no mitochondria, yet they do it

Bio · U3 · L1 · B · 14/24

About 36 ATP per glucose comes out

Bio · U3 · L1 · B · 22/24

Atoms are rearranged; none are created or destroyed

Bio · U3 · L1 · B · 23/24

Happens only in light

Bio · U3 · L1 · B · 04/24

Light energy becomes chemical energy in sugar

Bio · U3 · L1 · B · 13/24

Water goes in Bio · U3 · L1 · B · 15/24	Glucose goes in Bio · U3 · L1 · B · 09/24	Oxygen comes out Bio · U3 · L1 · B · 24/24	Runs day and night Bio · U3 · L1 · B · 11/24
Contains chlorophyll Bio · U3 · L1 · B · 16/24	Energy in glucose becomes energy in ATP Bio · U3 · L1 · B · 10/24	Found in plant *and* animal cells Bio · U3 · L1 · B · 08/24	Glucose comes out Bio · U3 · L1 · B · 07/24
Found in plant and algae cells only Bio · U3 · L1 · B · 12/24			

Deck C · Unit 3 Word Cards

Unit 3 · Energy and Matter
48 cards + this cover
Label code **Bio · U3 · L1 · C**
Envelope: **U3-C** · Set ____ of ____
Print single-sided on yellow cardstock. Count against this card before the set goes back in the bin.
Key: OneDrive › Unit 03 › Answer Keys

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photosynthesis
Bio · U3 · L1 · C · 32/48

chloroplast
Bio · U3 · L1 · C · 04/48

chlorophyll
Bio · U3 · L1 · C · 40/48

pigment
Bio · U3 · L1 · C · 10/48

glucose
Bio · U3 · L1 · C · 37/48

ATP
Bio · U3 · L1 · C · 24/48

ADP
Bio · U3 · L1 · C · 44/48

phosphate group
Bio · U3 · L1 · C · 01/48

cellular respiration
Bio · U3 · L1 · C · 45/48

mitochondrion
Bio · U3 · L1 · C · 12/48

aerobic
Bio · U3 · L1 · C · 31/48

anaerobic
Bio · U3 · L1 · C · 42/48

fermentation
Bio · U3 · L1 · C · 07/48

lactic acid
Bio · U3 · L1 · C · 03/48

reactant
Bio · U3 · L1 · C · 05/48

product
Bio · U3 · L1 · C · 30/48

metabolism
Bio · U3 · L1 · C · 46/48

conservation of matter
Bio · U3 · L1 · C · 18/48

autotroph
Bio · U3 · L1 · C · 29/48

heterotroph Bio · U3 · L1 · C · 08/48	**light intensity** Bio · U3 · L1 · C · 15/48	**indicator** Bio · U3 · L1 · C · 21/48	**cyanobacteria** Bio · U3 · L1 · C · 33/48
energy transformation Bio · U3 · L1 · C · 20/48	A substance that goes into a chemical reaction and is used up. Bio · U3 · L1 · C · 22/48	Energy changing form, such as light energy into chemical energy. Bio · U3 · L1 · C · 36/48	Atoms are rearranged in a reaction but never created or destroyed. Bio · U3 · L1 · C · 19/48
Light energy, carbon dioxide and water are used to make glucose and oxygen. Bio · U3 · L1 · C · 34/48	An organism that makes its own food, such as a plant (a producer). Bio · U3 · L1 · C · 06/48	Prokaryotes that photosynthesize with no chloroplasts. Bio · U3 · L1 · C · 23/48	How bright the light is; it changes the rate of photosynthesis. Bio · U3 · L1 · C · 14/48
The green pigment that captures light energy. Bio · U3 · L1 · C · 25/48	A substance that comes out of a chemical reaction. Bio · U3 · L1 · C · 41/48	A six-carbon sugar, $C_6H_{12}O_6$, that stores chemical energy. Bio · U3 · L1 · C · 38/48	Glucose and oxygen are used to make carbon dioxide, water and ATP. Bio · U3 · L1 · C · 28/48
What muscle cells make when they break down sugar without oxygen. Bio · U3 · L1 · C · 27/48	Breaking down sugar without oxygen, for only about 2 ATP per glucose. Bio · U3 · L1 · C · 17/48	The part that breaks off ATP when energy is released. Bio · U3 · L1 · C · 43/48	The organelle in plant and algae cells where photosynthesis happens. Bio · U3 · L1 · C · 13/48

Without oxygen.

Bio · U3 · L1 · C · 16/48

What ATP becomes after it loses one phosphate group.

Bio · U3 · L1 · C · 11/48

The molecule that carries energy a cell can spend right away.

Bio · U3 · L1 · C · 02/48

The organelle where most of a cell's ATP is made, using oxygen.

Bio · U3 · L1 · C · 48/48

A chemical, such as bromothymol blue, that changes color to show a change.

Bio · U3 · L1 · C · 47/48

With oxygen.

Bio · U3 · L1 · C · 39/48

A molecule that absorbs some colors of light and reflects others.

Bio · U3 · L1 · C · 09/48

All the chemical reactions that happen in a living thing.

Bio · U3 · L1 · C · 35/48

An organism that must eat other organisms for food (a consumer).

Bio · U3 · L1 · C · 26/48